

A Multi-Layer Secure Authentication Framework Using Dynamic QR Codes and Geofencing for Real-Time Attendance Verification

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Abstract

Attendance management is an essential administrative process in educational institutions and plays a vital role in the observation of attendance records, maintenance of academic discipline, and decision making. Conventional methods of attendance management such as manual roll calls and pen and paper based registration process are cumbersome, error-prone, and susceptible to proxy attendances [1, 2]. Though there have been various suggestions on automatic attendance management systems employing RFID, Biometrics, QR code and location-based techniques, most of the systems need special hardware, more deployment costs or less analytical support [3, 21].

The proposed approach in this paper is **A Multi-Layer Secure Attendance Management Framework**, which is a secure and intelligent attendance management system, which makes use of **dynamic QR code verification**, **geofencing-based location verification**, and **attendance analysis** all together in a single web-based system. In the proposed system, periodic refreshment of QR codes reduces the instances of QR sharing and proxy attendance, and geofencing guarantees that attendance is recorded only if the student is present inside the specific radius from the classroom [2, 5].

In order to conduct an assessment of the efficiency of the proposed system, a prototype was created and tested through an academic environment, which involved the following number of participants and processes – **75 students, 15 teachers, 1 administrator, 184 attendance periods, and 2,745 attendance records**. The result of experimental evaluation reached the accuracy of attendance verification **98.89%**, proving that there is a considerable increase in efficiency of attendance record, attendance reliability, and user satisfaction as opposed to traditional attendance methods.

Therefore, the results suggest that Smart Attendance Guard offers a highly efficient, scalable and practical solution to the problem of modern attendance management in education sector.

Keywords: Attendance Management System, Dynamic QR Code, Geofencing, Educational ERP, Proxy Attendance Detection, Real Time Attendance Monitoring, Attendance Analysis, Attendance Verification.

1 Introduction

Educational institutions are incorporating digital technology into their institutions for better academic management, process efficiency, and effective data handling. Of all these processes, attendance management becomes significant in the sense that it enables tracking of attendance of students, enforcement of policy, and decision making in academics. The conventional method of manual attendance recording and use of registers for recording the attendance is not only tedious but prone to errors and proxy attendance [1, 2].

Various automatic attendance systems have been developed through RFID, biometric system, QR code system, and GPS-based attendance system in order to automate attendance recording. Despite these attempts to make the process easier, various problems such as dependency on hardware, privacy issues, screenshots, replay attack, costliness, and lack of analytics capability persist [3, 5, 21]. Thus, there is a need for developing an attendance management system that is capable of providing security, low cost, scalability, and deployability.

To overcome the above shortcomings, this paper introduces **A Multi-Layer Secure Attendance Management Framework**, which is an attendance management system based on the Web with **dynamic QR code authentication, location validation based on geofencing, and attendance analytics**. Dynamic QR codes are used periodically in this paper to overcome the threat of QR code sharing and QR replay attack, whereas geofencing will ensure attendance is recorded if the student is physically present in a predefined classroom space [2, 5].

The main contributions of this paper can be stated as follows:

1. Development of a secure attendance management framework utilizing Dynamic QR code authentication, geofencing, and JWT-based role-based authentication.
2. Usage of time-sensitive Dynamic QR code generation process to minimize QR replay attacks and QR sharing.
3. Use of geofencing-based classroom verification using Haversine distance calculation method to validate the physical presence of students.
4. Role-based dashboards designed for attendance management, monitoring, reporting, and attendance analytics.

5. Experimental evaluation using **75 students, 15 faculty members, one administrator, 184 attendance sessions, and 2,745 attendance logs.**
6. Mathematical modeling, confusion matrix analysis, hypothesis testing, and comparative evaluation providing an attendance verification accuracy of **98.89%**.

The rest of this paper is arranged as follows. Section II provides the related works, Section III provides the research gap analysis, Section IV explains the novelty of the proposed work, Section V explains the system design and architecture, Section VI explains the materials and methods, Section VII explains the mathematical model, Section VIII explains the experimental setup, Section IX provides the evaluation metrics, Section X provides the dataset description, Section XI provides the experimental results and discussion, while the subsequent sections provide statistical analysis, comparative evaluation, conclusions, and future research directions.

2 Related Works

The traditional attendance management system has been transformed to some extent from paper-based traditional attendance records into intelligent electronic systems, which help to make processes more efficient and more accurate and monitor attendance easily. The recent study revolves around the automated attendance system through QR codes, RFID technology, geofencing, AI of things (AIoT), and cloud-based educational platforms, overcoming issues faced by manual attendance systems.

The manual attendance systems are easy to design but take up valuable class time and prone to human errors and proxy attendance issues. Many systematic literature reviews have discussed these issues and the importance of automated attendance systems, providing real-time monitoring and authentication mechanism capability [1, 8, 20].

Attendance systems using QR code are one of the most commonly used approaches owing to low cost, ease of deployment, and usability with smartphones. There are many studies conducted showing that the QR code attendance system takes less time in attendance recording than the traditional method [4, 7]. However, static QR code-based attendance system is vulnerable to screenshot attack, replay attacks, and proxy attendance. In order to solve such problems, a

Dynamic QR Authentication system has been suggested in recent literature, which involves the regular regeneration of QR codes in order to invalidate old issued tokens and make attendance more authentic [2, 5, 6, 9, 10, 13, 15]. While such solutions significantly increase the level of security, they do not usually include the validation of user presence in the lecture room.

However, location-aware attendance systems use GPS technology along with the geofencing technique, which allows confirming the fact of the user's presence within defined institutional boundaries before taking attendance. Geofencing-based attendance systems decrease the risks of remote attendance fraud, limiting the attendance submission to authorized locations [3, 11]. Recent literature has also combined QR authentication with geofencing for providing an additional step of verification by means of token and location checking at once [2, 5, 6, 10, 12–14]. Still, geolocation methods are highly vulnerable to GPS errors and location spoofing without the use of authentication systems.

The other technology that has been extensively researched in relation to attendance automation due to the fast user identification capabilities is the RFID technology [21]. Existing attendance systems using RFID technology offer higher efficiency and reduced recording time as compared to manual attendance systems [21]. Moreover, modern RFID-based systems have added various other sensors and the two-factor authentication to counter the problem of proxy attendance [16, 22]. However, despite these innovations, RFID systems still require dedicated hardware infrastructure and are expensive to deploy. Moreover, they are susceptible to the issue of card sharing.

The recent developments in the field of AIoT technology and smart classrooms have extended the scope of attendance management. AIoT-based attendance management systems involve the use of IoT devices, cloud computing services, intelligent monitoring, and real-time analytics to better manage the institution and help with decision-making in education [17, 19]. In a similar way, smart classrooms have included behavioral analysis and emotion-aware monitoring in order to ensure classroom engagement and better evaluation of learning process [18]. Nevertheless, many intelligent attendance management systems require dedicated hardware.

In conclusion, the existing studies have made considerable advancements in automating the attendance process; yet there is no study that manages to incorporate all of the above-

mentioned components, such as dynamic QR code authentication, geofence-based localization of user locations, secure access control based on roles, attendance analysis, cost-effectiveness, scalability, and Web-based implementation into one approach. The shortcomings discussed above provide a rationale for developing the Smart Attendance Guard.

3 Research Gap Analysis

According to the literature review, even though current attendance management systems automate the process, they have a number of weaknesses. In the case of manual system, the process is ineffective and allows for proxy attendance; RFID-based approach requires additional hardware to be installed; static QR code can be captured on the screen; geolocation systems cannot validate a user's identity alone; and biometric system raises issues related to privacy and implementation.

The research gaps are as follows:

- Multi-tiered attendance authentication is absent.
- No proper defense mechanism to stop proxy attendance.
- Secured attendance solution has expensive implementation costs.
- Little availability of attendance analytics and real-time monitoring.
- Role-based management is insufficient for various parties.

In order to overcome these shortcomings, this research suggests **A Multi-Layer Secure Attendance Management Framework** that combines dynamic QR code authentication, geofencing authentication, JWT-based role management, and attendance analytics in a web-based system design. The suggested methodology makes attendance verification more reliable without sacrificing low costs and easy deployment. From the comparative analysis shown in Table 1, it can be clearly seen that the current attendance management system only takes care of one or two factors of attendance validation. The manual attendance system lacks automation, RFID attendance system remains vulnerable to the sharing of cards, static QR system is prone to QR

Table 1: Comparison of Existing Attendance Management Approaches

Attendance System	Dynamic QR	Geofencing	Analytics	Proxy Prevention	ERP Integration
Manual Attendance	×	×	×	×	×
RFID-Based System	×	×	Limited	Limited	Limited
Static QR Attendance	×	×	Limited	×	Limited
Geolocation Attendance	×	✓	Limited	Limited	Limited
Proposed Framework	✓	✓	✓	✓	✓

forwarding/replay attack, and geolocation alone does not verify any authentication of its own. Despite the significant advancements that have been made in terms of attendance automation via QR authentication, RFID, geofencing, and artificial intelligence-enabled monitoring, there are no existing frameworks which offer comprehensive attendance security. Although dynamic QR-based attendance authentication techniques successfully counter replay attacks, they are unable to independently confirm the physical presence of users. In the same manner, although geofencing can be used as an attendance authentication technique, it is subject to inaccurate readings by the GPS and can easily be bypassed using location spoofing. RFID-based systems, on the other hand, need specialized hardware and, therefore, are costly to deploy and maintain. Moreover, most intelligent attendance systems offer little analytical functionality that would be useful in decision making at the institutional level [2, 5, 15, 20, 21]. This problem necessitates the development of the proposed Smart Attendance Guard, which integrates dynamic QR authentication, geofencing, authorization through JWT, and attendance analytics in a web-based platform.

4 Novelty of the Proposed Authentication Framework

The proposed **A Multi-Layer Secure Attendance Management Framework** is an attendance management system that is based on the use of Dynamic QR code authentication, geofencing-based location verification, attendance analytics and role-based access control using JSON Web Tokens into a single platform for web-based education systems. Though several attendance management systems in the past have included some or other method of QR codes, geolocation, RFID, or biometric, but mostly a single authentication factor was being used.

First of all, the key innovation of the proposed model is in the use of the **multi-level atten-**

dance verification architecture that requires any attendance application to go through three verification levels: authentication of the authorized user, validation of the dynamic QR token at a particular moment of time and validation of the classroom location based on geofencing.

In contrast to the traditional static QR token generation mechanisms, in the proposed Dynamic QR module unique QR tokens will be generated every seven seconds according to session parameters, timestamps and random nonces. Thus, the proposed mechanism helps to prevent possible problems associated with QR forwarding and reuse without increasing the complexity of the attendance process. Additionally, geofencing module will verify the presence of the student inside the predefined classroom perimeter based on the Haversine distance model.

Apart from that, the proposed framework provides role-based dashboard for the teacher, student and administrator, real-time attendance verification and reporting, auditing and analytical visualizations of attendance statistics.

Furthermore, the framework was developed as a full-fledged web-based prototype utilizing state-of-the-art open-source technologies and analyzed based on mathematical modeling, functional testing, confusion matrix evaluation, statistical hypothesis testing, and comparative performance assessment. The obtained experimental results show that the developed multi-layer framework for attendance verification provides high accuracy of attendance verification while being highly cost-effective, scalable, and easy to adopt.

The main contribution of this research lies not only in the implementation of the concept of dynamic QR codes and geofencing, but in their successful integration into an efficient, multi-layer attendance management system.

5 System Design and Architecture

5.1 System Overview

The **A Multi-Layer Secure Attendance Management Framework** framework is an internet-based attendance management framework that ensures secure and efficient attendance verification for educational institutions. The framework makes use of Dynamic QR code authentication, geofencing location verification, JWT authentication mechanism for role-based authentication,

and attendance analytics in real-time within a multi-layered framework.

Three types of users have been taken into consideration within this framework – the teachers, the students, and the administrators. Each of these users has a specific dashboard allocated to them depending on their job responsibility. Teachers start an attendance session, generate dynamic QR code, monitor attendance in real-time, and generate attendance reports, while the students authenticate themselves and scan QR code and make attendance requests, respectively. The process of attendance verification is based on a sequential workflow of several layers. First of all, verified users authenticate to the system by means of the JWT-based authentication process. During the attendance check process, the teacher creates a Dynamic QR code which is valid within a specified period. The location of the student is retrieved via the Browser Geolocation API after the QR code scan and is checked against the specified classroom location with the help of the Haversine distance formula. The attendance is registered only in case of successful verification of user authentication, dynamic QR, and geofencing.

The suggested architecture is based on the modular layered approach including such layers as Presentation Layer, Application Layer, and Data Layer. Such structure of the system allows enhancing its maintainability, scalability, security, and extensibility. Figure 1 shows the general architecture of the proposed Smart Attendance Guard system.

5.2 System Modules

The proposed Smart Attendance Guard system consists of connected functional modules performing different tasks but communicating via RESTful APIs and the common database.

Teacher Module: The Teacher Module facilitates creating of attendance sessions, creation of Dynamic QR codes, tracking attendance status of the students in real-time, and creating attendance report. The dynamic nature of QR token refreshes itself at regular intervals during an attendance session to avoid sharing of QR code images and re-playing attack. It further synchronizes attendance details with the backend server.

Student Module: The Student Module enables authenticated students to participate in the attendance session, scanning Dynamic QR codes, and submitting attendance request. Prior to submission of the attendance request, geographical location details of the student are fetched

by the Browser Geolocation API and sent to the backend for geofencing authentication. Students can see their attendance record, attendance percentage, and subject wise attendance record through personalized dashboards.

Administrator Module: The Administrator Module helps in central management of institutional data such as user accounts, academic schedule, attendance record, and system settings. Administrators can analyze attendance trends, generate reports, and audit logs, and manage institutional attendance activities through administrative dashboards. **Authentication Module:** This module uses JWT and Bcrypt password hashing to provide a secure mechanism for user authentication and authorization. The module authenticates each request prior to making any attendance requests and ensures that the request is authorized for attendance tasks based on the role of the user.

Dynamic QR Module: In the Dynamic QR Module, unique and time-sensitive QR tokens are generated for each attendance instance. The QR tokens include encrypted attendance session information that updates after a certain period of time so that no repeated usage of QR token takes place.

Geofencing Module: In the Geofencing Module, geographical coordinates of students are checked against the predefined geographical coordinates of the class room using Haversine distance formula. Attendance requests outside the authorized attendance radius are not considered as valid.

Attendance Analytics Module: The Attendance Analytics Module analyzes the attendance data and generates statistical analysis, attendance percentage, subject-wise report, branch-wise analysis, and institution-wide attendance trends. The generated reports will help the teachers and administration keep track of academic performance and attendance compliance.

The collaboration between the above functional modules enables the creation of a secure, scalable and efficient attendance management system that would reduce proxy attendance and provide real-time monitoring of attendance and analysis of the institution.

5.3 Multi-Layer Attendance Verification Process

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5.4 Attendance Analytics

The Attendance Analytics Module is responsible for processing attendance records into useful statistics to monitor the learning process, help in decision-making and manage attendance. Unlike traditional attendance systems which only log attendance status, the proposed framework processes attendance records continually to provide real-time analytical reports for the teacher, students, and administrators. The analytics module automatically generates attendance percentages, attendance statistics by subjects, attendance summaries by classes, attendance comparisons by branches and institution-wide attendance trends using the data entered in the attendance module into the central database. The reports generated are constantly updated after each successful transaction, thus ensuring that users have access to the latest attendance data without generating reports manually.

The teachers are able to keep track of the student attendance, identify any abnormal attendance patterns and also generate reports for the attendance of particular classes. The students on their side are able to view their attendance percentage, attendance record and attendance performance per subjects on the personalized dashboard provided. The administrators will be able to view institution-wide attendance summaries, attendance comparisons between departments, audit trails and statistical reports that facilitate administrative decision making.

Apart from the descriptive statistics, the analytics module also generates graphs showing the trends in attendance as well as any attendance patterns that have been experienced historically. This is done with the aim of identifying any students who have low attendance levels.

With the implementation of attendance verification along with real-time analytics, the framework will not only make attendance recording automatic but will also help in transforming attendance data into institutional intelligence. Thus, with the help of analytics module, efficiency in operations and reduction in administrative burden would be facilitated.

5.5 Threat Model and Security Analysis

Some of the major security threats that can affect the operation of attendance management systems in educational institutes are proxy attendance, QR code sharing, replay attack, system access without authorization, duplicate attendance submission, and submission of attendance outside classroom area. Such security threats decrease the validity of attendance reports and could affect institutional decision making process.

The proposed Smart Attendance Guard framework incorporates multi-layered security architecture using JWT-based authentication, Dynamic QR authentication and Geofencing verification to deal with the aforementioned security threats. Instead of using a single authentication method, attendance is permanently marked after passing all the stages of authentication. Table 2 provides a summary of major security threats taken into account while designing the system along with their respective mitigations in the designed framework. The threat model above

Table 2: Threat Model and Security Mitigation

Security Threat	Mitigation in the Proposed Multi-Layer Authentication Framework
Proxy Attendance	The combination of JWT authentication, dynamic QR verification, and 50-meter geofencing ensures that both user identity and physical presence are verified before attendance is recorded.
QR Screenshot Leakage	Dynamic QR codes are regenerated every 7 seconds , preventing the reuse of previously captured QR codes.
Replay Attack	The server validates the expiration time of each dynamic QR token, preventing replay attacks and reuse of expired QR codes.
Unauthorized System Access	JWT authentication together with role-based access control (RBAC) restricts system access to authenticated students, faculty members, and administrators.
Attendance Outside Authorized Area	Attendance is accepted only when the user's current location falls within the predefined 50-meter geofencing boundary.
Duplicate Attendance Submission	Server-side validation ensures that only one attendance record can be submitted for each attendance session.
Attendance Record Manipulation	Authenticated communication, secure database operations, and access control mechanisms prevent unauthorized modification of attendance records.

indicates that the proposed Smart Attendance Guard system uses a layered security approach to enhance attendance authenticity. The proposed system differs from traditional attendance systems in that they use a single approach of authentication whereas the proposed one incor-

porates multiple approaches of JWT Authentication, Dynamic QR Validation, and Geofencing Verification into one authentication approach. Dynamic QR codes refreshed every **7 seconds** make it difficult for forwarding and replay attacks on the system to happen, and the **50 meters geofencing** distance makes sure that attendance can only be marked if the user is physically present in the classroom.

5.6 System Architecture Diagram

Figure 1 represents the overall architecture of the proposed Smart Attendance Guard framework. The system follows the architectural design where the User Layer, Presentation Layer, Application Layer, Data Layer, and External Services are divided into separate layers, providing separation between user interaction, business logic, data management, and external services.

User Layer: The User Layer contains three types of users: teachers, students, and administrators. Teachers start the attendance session, create Dynamic QR codes, and check the status of attendance in real-time. Students perform authentication, scan Dynamic QR codes, and request for attendance. Administrators control the work of the institution, manage users' accounts, generate reports, and analyze attendance.

Presentation Layer: The Presentation Layer contains responsive web interfaces, which were developed using React.js and Tailwind CSS. Different dashboards are designed for teachers, students, and administrators depending on the roles of each of them.

Application Layer: This Application Layer contains all business logic of the proposed framework. JWT-based authentication, Dynamic QR generation, QR validation, geofencing verification, attendance processing, analytics generation, real-time communication via Socket.io, and REST API services form the Application Layer which controls the entire attendance verification process and stores attendance entries only after the successful execution of all the validation steps.

Data Layer: The Data Layer keeps all the persistent data used by the system and includes user profile data, attendance data, attendance sessions data, class schedule data, attendance analytics data, and administrative audit log data. The backend database for the attendance management system is SQLite which provides reliable storage and data retrieval during the

process of attendance management.

External Services: These are the external services integrated into the proposed attendance management framework and include the Browser Geolocation API for obtaining geographical coordinates and the email notification service for sending out attendance-related data. These are the services which increase the capabilities of the attendance management system without increasing its implementation complexity. All together, the layered approach allows achieving safe communication between the system modules along with modular software development, attendance verification, centralized data management, and real-time analytical reporting. The defined roles of each architectural layer make it scalable, maintainable, and easy to integrate with cloud-based educational management systems.

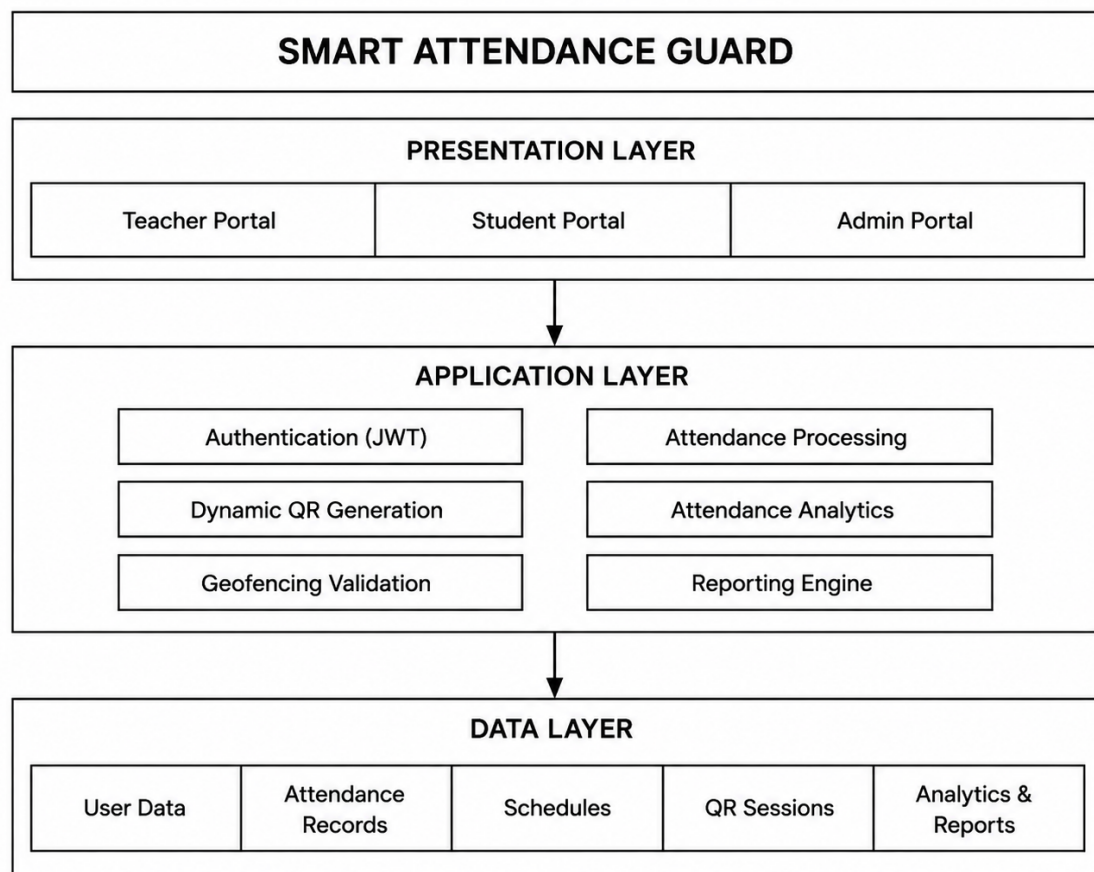


Fig. 1. Layered architecture of the Smart Attendance Guard system.

Figure 1: System Architecture of the Proposed Smart Attendance Guard Framework

6 Materials and Methods

6.1 Development Environment

The prototype of the Smart Attendance Guard framework was developed utilizing React 18 and Tailwind CSS on the frontend side, Node.js with Express.js on the backend services side, SQLite as a database management solution, and Socket.io for real-time communications. The user authentication is realized using JSON Web Tokens (JWT) with bcrypt password hashing, whereas Browser Geolocation API is used for location verification.

6.2 Technology Stack

Table 3 shows the technology stack which has been used during the development of the system.

Table 3: Technology Stack for Smart Attendance Guard System

Layer	Technology
Frontend	React 18, Vite, Tailwind CSS
Backend	Node.js, Express.js
Database	SQLite
Authentication	JWT, Bcrypt
Real-Time Communication	Socket.io
QR Code Processing	QR Code Generation Libraries
Geolocation Services	Browser Geolocation API
Development Tools	Visual Studio Code, Git

6.3 Design Rationale

The suggested multi-layered authentication approach was developed to address the issue of attendance verification in educational institutions. The selection of technology stack and architectural decisions were based on security criteria, simplicity of implementation, efficiency, and scalability.

For frontend development React.js was used due to its component architecture and quick rendering, as well as responsive design. For backend, Node.js and Express.js were selected as tools that allow implementing asynchronous requests and communication of the client applica-

tion with other resources. As the database management tool SQLite was used because of its simple architecture and easy configuration.

The dynamic QR code authentication was chosen in order to prevent proxy attendances because of the time-bound authentication token that expires after certain period. As compared to the static QR code, it provides much lower chances of usage of screenshots for unauthorized purposes. Dynamic QR code authentication with geofencing was used as an additional step in the process of confirming the physical presence of users in the approved attendance location. The Haversine distance computation algorithm was applied in order to correctly compute whether users were present within the defined geographical boundary.

Role-Based Access Control (RBAC) was introduced in order to make sure that students, lecturers, and administration get access rights appropriate for their functions. JSON Web Tokens (JWT) were applied for session management and secured interaction between frontend and backend components.

In general, the combination of dynamic QR authentication, geofencing, authentication, and role-based authorization creates a robust multi-layered security system which can enhance the attendance security and be both computationally efficient and user-friendly.

6.4 Architecture of the Frontend

The frontend uses a component-based React architecture that allows creating dedicated dashboards for teachers, students, and admins. The frontend communicates with the backend using RESTful APIs, and React. React.js has been used for developing the frontend due to its component-based nature and virtual DOM rendering technique .

6.5 Backend Architecture

The back-end is designed based on Node.js and Express.js using a layered architecture comprising an authentication layer, business logic layer, and database access layer. RESTful APIs will be used to handle attendance requests, validate QR tokens, check geofencing rules, and manage attendance synchronization using Socket.io. Node.js and Express.js were used to develop the backend services, due to their ability to handle asynchronous requests and RESTful APIs .

6.6 Database Schema

The database schema consists of tables containing user information, attendance records, attendance sessions, schedules, analysis, and audit logs. Relationships between primary and foreign keys provide data consistency. Normalization will reduce redundancy and enable efficient attendance retrieval.

Table 4: Database Schema

Table Name	Purpose
Users	Stores authentication and profile information for students, teachers, and admin.
Attendance	Stores attendance records created in attendance sessions.
Sessions	Stores details about attendance sessions, QR tokens, times stamps, and sessions status.
Schedule	Stores classes' schedules and timetable information.
Analytics	Stores attendance summaries and statistics reports.
AuditLogs	Stores records of attendance changes and management actions.

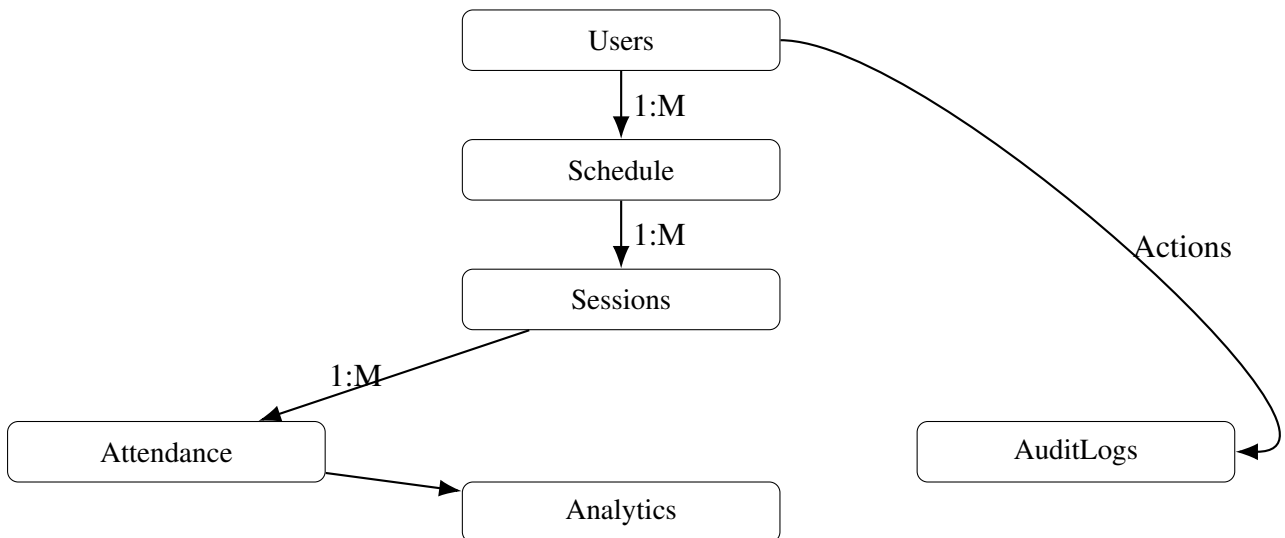


Figure 2: Entity Relationship Overview of the Proposed Database

The design of the relational database system guarantees efficient storage of attendance data, proper management of information, analysis capabilities, and scalability of the system for future implementation in any institution. SQLite has been chosen based on its lightweight nature, portability, reliability, and good transaction management for education-based web applications .

6.7 Authentication

Using JWT authentication enables secure logins and role-based authorization. Passwords will be stored encrypted by using Bcrypt algorithm, and authenticated users will. This process of authentication is done by following the web authentication and digital identity management best practices .

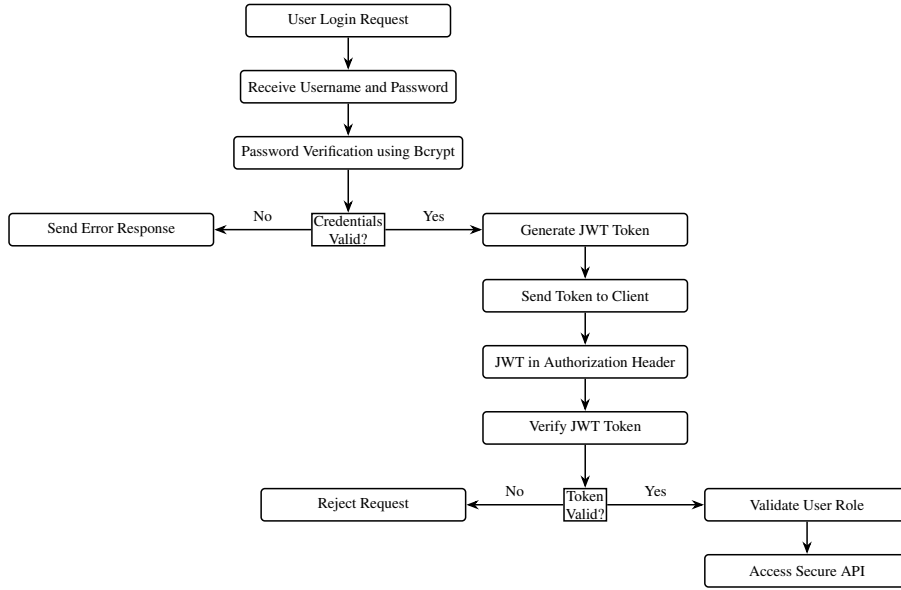


Figure 3: JWT Authentication Flow

Integration of JWT authentication, bcrypt, and role based authorization allows building an efficient and reliable security solution that can help to

6.8 Dynamic QR Algorithm

Each attendance session produces a unique QR token based on session identifier, teacher identifier, timestamp, and random nonce. QR codes are refreshed automatically after every seven seconds in order to avoid any possibility of replay attacks and QR code reuse. The processing of attendance happens only after the verification of token.

Generation of unique QR token can be symbolically represented as

$$QR_{\text{token}} = H(\text{SessionID} \parallel \text{TeacherID} \parallel \text{Timestamp} \parallel \text{RandomNonce}) \quad (1)$$

where,

- **SessionID** is a unique identifier for an attendance session.
- **TeacherID** is a unique identifier for the teacher giving the class.
- **Timestamp** is the token creation timestamp.
- **RandomNonce** is a randomly generated value to ensure uniqueness of QR tokens.
- $H(\cdot)$ is a secure cryptographic hash function that produces a unique attendance token.

Use of timestamps and nonces ensures that QR tokens produced are always unique even if the same teacher starts multiple attendance sessions. Moreover, the validation process done server side ensures that the QR tokens cannot be modified and reused.

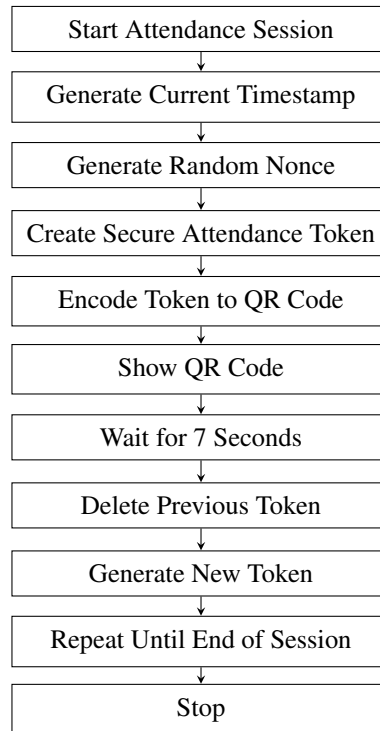


Figure 4: Dynamic QR Code Generation Algorithm

The above-mentioned dynamic QR generation technique offers several security benefits over the static QR attendance systems currently in use. First, refreshing of the QR codes regularly minimizes the possibility of attendance fraud through the screenshot method. Second, server-side validation eliminates the possibility of acceptance of expired or modified QR codes. Third, using the combination of dynamic QR and geofencing creates an attendance authentication process with multi-level security features without much overhead.

The security level is increased compared to traditional static QR techniques due to higher protection against possible replay attacks and enhanced scalability without any need for additional hardware setup. Therefore, the Dynamic QR module becomes the first step in the Smart Attendance Guard system.

6.9 Geofencing Validation

The browser Geolocation API gets the position of the student and Haversine formula calculates the distance from the classroom. The attendance is validated if the distance is less than the specified geofencing radius, which guarantees the presence of the student in the classroom. The distance between the teacher and student is measured by

$$d = 2R \sin^{-1} \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right) \quad (2)$$

where

- R represents the Earth's radius (6371 km),
- ϕ_1, ϕ_2 denote the latitudes of the teacher and student,
- λ_1, λ_2 denote the corresponding longitudes,
- $\Delta\phi = \phi_2 - \phi_1$,
- $\Delta\lambda = \lambda_2 - \lambda_1$.

After getting the distance, the program then compares the calculated distance to the geofencing threshold set to **50 meters**. The attendance will be allowed only if the student is within this radius. In case the distance is more than the threshold, attendance will be denied irrespective of the QR code's validity.

The chosen threshold of 50 meters ensures a good balance between securing the attendance process and usability of the same. This allows for some GPS errors and at the same time guarantees that attendance is allowed only to those students who are physically present around the classroom.

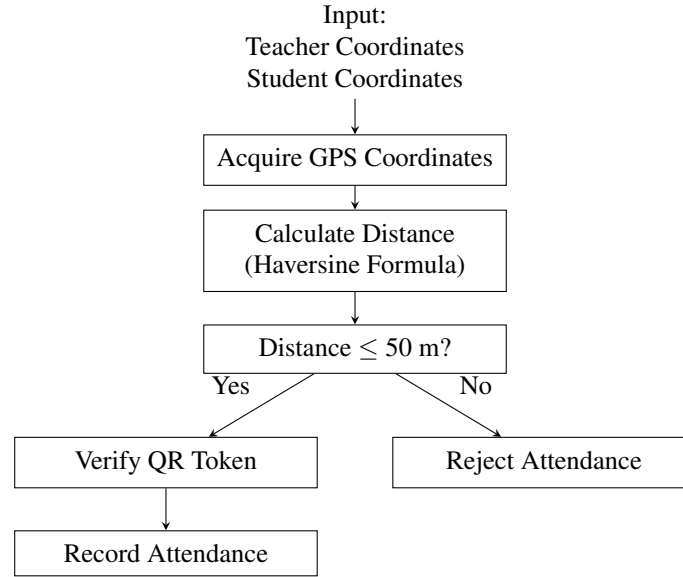


Figure 5: Geofencing Validation Algorithm

The discussed geofencing technique would considerably improve the authenticity of attendance checks due to the implementation of both dynamic QR verification and location-based authentication. If a third-party acquires a legitimate QR token, they would still have to be present within the specified attendance boundary in order to log their attendance. Thus, the proposed framework based on Dynamic QR and geofencing provides multiple layers of protection against fraudulent attendance logging and location spoofing without increasing the complexity of the process.

6.10 Computational Complexity Analysis

The Smart Attendance Guard system has been created with lightweight mechanisms of authentication and validation in order to make the attendance check process secure but free of unnecessary computations. The computational complexity of the key operations is listed in Table 5.

Since the framework performs Dynamic QR generation, JWT verification, geofencing validation, and attendance recording in sequential order per every attendance request, the complexity of the above operations does not depend on the number of users. Therefore, the framework can be used for real-time attendance management in educational settings. The analysis of the complexity shows that all operations associated with attendance verification in real time, in-

Table 5: Computational Complexity of Major Operations

Operation	Complexity
JWT Authentication	$O(1)$
Dynamic QR Generation	$O(1)$
QR Validation	$O(1)$
Geofencing Verification	$O(1)$
Attendance Recording	$O(1)$
Attendance Analytics Retrieval	$O(n)$
Report Generation	$O(n)$

cluding JWT authentication, Dynamic QR generation, QR validation, geofencing verification, and attendance recording, perform in constant time. Therefore, the proposed framework is able to authenticate attendance requests efficiently without computational delay.

Operations for the purpose of attendance analytics and report generation require processing historical attendance records and thus have linear complexity in terms of the number of attendance records stored. Such operations are conducted asynchronously and do not impact the process of attendance recording in real time.

7 Mathematical Model

The Smart Attendance Guard framework conducts attendance verification with the help of multi-stage authentication procedure, which includes user authentication, dynamic QR code validation, and geofencing verification. The record of attendance is created only when all security verifications are performed successfully.

Let

- U_v be the variable denoting user authentication validation,
- QR_v be the variable denoting dynamic QR code validation,
- G_v be the variable denoting geofencing validation.

Each of the above validation variables is binary and defined as

$$U_v, QR_v, G_v \in \{0, 1\}$$

where

$$1 = \text{Validation Successful}$$

$$0 = \text{Validation Failed}$$

The attendance authorization function is expressed as

$$A = U_v \times QR_v \times G_v$$

where

$$A = \begin{cases} 1, & \text{Attendance Recorded} \\ 0, & \text{Attendance Rejected} \end{cases}$$

Thus, attendance is recorded only if

$$U_v = QR_v = G_v = 1$$

Otherwise,

$$A = 0$$

The mathematical equation thus ensures that any unauthorized person, expired QR code, or student who is outside the classroom perimeter cannot register their attendance.

7.1 Geofencing Distance Calculation

The geographical distance between the teacher and the student is calculated through the use of the Haversine formula. Let

$$(\phi_1, \lambda_1)$$

represent the teacher's geographical coordinates and

$$(\phi_2, \lambda_2)$$

represent the student's geographical coordinates.

The distance is calculated as

$$d = 2R \arcsin \left(\sqrt{\sin^2 \left(\frac{\phi_2 - \phi_1}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\lambda_2 - \lambda_1}{2} \right)} \right)$$

where

- $R = 6371$ km represents the Earth's radius,
- d represents the geographical distance.

Attendance is authorized only when

$$d \leq 50 \text{ meters}$$

Otherwise, attendance is rejected.

7.2 Attendance Accuracy

The attendance verification accuracy is computed using the confusion matrix.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

where

- TP = True Positive
- TN = True Negative

- FP = False Positive
- FN = False Negative

7.3 Precision

Precision measures the proportion of correctly accepted attendance requests among all accepted requests.

$$Precision = \frac{TP}{TP + FP}$$

7.4 Recall

Recall measures the proportion of correctly authenticated attendance requests among all actual valid attendance requests.

$$Recall = \frac{TP}{TP + FN}$$

7.5 F1-Score

The harmonic mean of Precision and Recall is computed as

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

The F1-Score gives a balanced measure of the performance of attendance verification by taking into account both the false acceptance rate and the false rejection rate.

7.6 Proxy Prevention Rate

Proxy prevention can be measured with the help of the Proxy Prevention Rate (PPR).

$$PPR = \frac{Detected\ Proxy\ Attempts}{Total\ Proxy\ Attempts} \times 100$$

The higher value suggests increased resistance to fraud attempts in terms of attendance.

7.7 Authentication Workflow

The flow for processing the attendance includes the steps such as teacher authentication, dynamic QR code generation, student authentication, QR scanning, geofencing verification, logging of attendance, and real-time synchronization. Any invalid QR code, duplicate request, and geofencing violation is immediately dropped from processing.

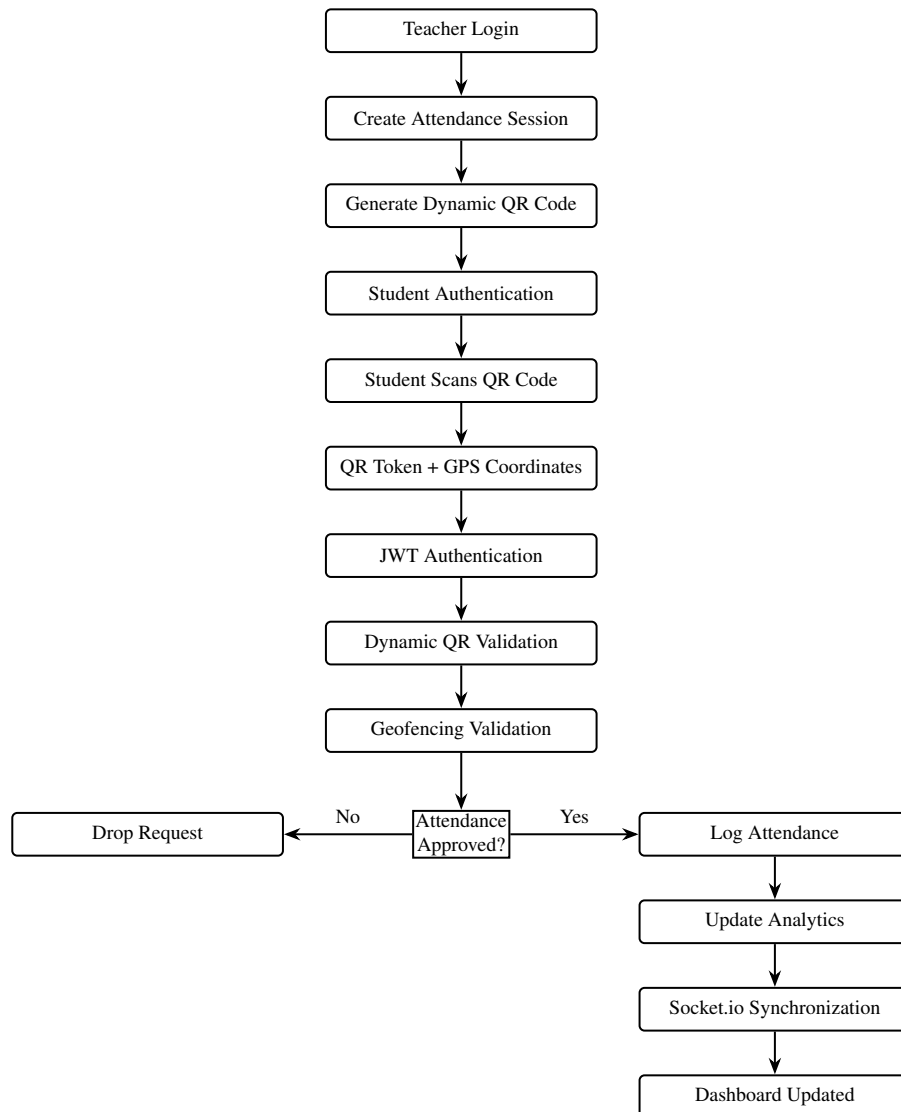


Figure 6: Attendance Recording Workflow

The suggested workflow for attendance recording is designed to build a secure and efficient pipeline of attendance verification through authentication, dynamic QR verification, geofencing verification, and real-time synchronization. Such a workflow reduces possibilities for proxy attendance, decreases human intervention and ensures attendance record generation at the successful completion of all verification steps. As a result, the suggested framework is a reliable

and scalable solution for attendance management in modern educational institutions.

8 Experimental Setup

8.1 Experimental Environment

A prototype of the A Multi-Layer Secure Attendance Management Framework was tested in a controlled academic environment to assess its effectiveness in solving attendance management issues. The prototype was installed on a local server and developed in React 18, Node.js, Express.js, SQLite, and Socket.io. For testing purposes, desktop computers and Android phones with Chromium-based browsers were used.

In attendance sessions, there was performed authentication, generation of dynamic QR, scanning QR, geofencing verification, attendance recording, and synchronization. Apart from usual attendance cases, such experiments as attendance with an expired QR, attendance duplication, invalid authentication, and attendance request from outside the authorized geofence area were done.

8.2 Participants

The experiment comprised 91 users divided into 75 students, 15 faculty members, and 1 administrator. The number of students was evenly divided among five different academic streams to measure the scalability of the framework under various academic conditions.

Attendance sessions were conducted by the faculty members using the Teacher Portal, and attendance was noted by the students using the Student Portal. Attendance records at an institutional level were monitored and analyzed by the administrators.

8.3 Attendance Sessions

A total of 50 attendance sessions were held in the experimental evaluation. These sessions entailed dynamic QR code generation, QR scanning, geofencing validation, and attendance recording. The experiments comprised successful attendance submissions as well as abnormal

Table 6: Comprehensive Participant Distribution

Participant Category & Sub-Group	Count	Percentage (%)
Students (80.65% of Total Users)	75	100.00
Computer Science and Engineering (CSE)	15	20.00
Electronics and Communication Engineering (ECE)	15	20.00
Electrical and Electronics Engineering (EEE)	15	20.00
Mechanical Engineering (MECH)	15	20.00
Civil Engineering (CIVIL)	15	20.00
Faculty Members (17.20% of Total Users)	15	100.00
Teaching Faculty	15	93.75
Head of Department / Academic Coordinator	1	6.25
Administrators (2.15% of Total Users)	1	100.00
Total System Users	91	-

scenarios like expiration of QR codes, duplicated attendance attempts, and geofence violations.

9 Evaluation Metrics

Performance of the system was analyzed through the following metrics:

- Accuracy of attendance recording
- Success rate of QR validation
- Geofencing validation accuracy
- Time of attendance recording
- User satisfaction
- System reliability

9.1 Attendance Recording Efficiency

Attendance recording efficiency reflects the time taken on average to complete the process of attendance recording starting from the beginning of the attendance session and ending at successful attendance recording confirmation. This metric assesses the speed of the proposed system

and its suitability for classroom use where attendance recording should take little instructional time.

9.2 Attendance Accuracy

Attendance accuracy is the ratio of correct attendance entries against the total number of attendance entries made through the proposed attendance management framework during experimentation. Attendance accuracy is used to determine the accuracy of attendance recording, and the causes of erroneous attendance records due to invalid authentication, outdated QR codes, or violation of geofencing conditions.

9.3 Proxy Prevention Effectiveness

Proxy prevention effectiveness is used to measure the ability of the proposed security framework to prevent proxy attendance submissions. It is measured in terms of percentage of unauthorized attendance attempts that are prevented by the combined use of JWT authentication, dynamic QR codes, and geofencing condition checks.

9.4 User Satisfaction

The user satisfaction with the proposed attendance management framework was measured separately for students and faculties to assess the usability and acceptance of the proposed solution. Users rated several features of the proposed solution including user interface, ease of attendance submission, response time, and other relevant parameters. The gathered feedbacks can help to understand the usability of the proposed attendance management framework.

9.5 Effectiveness of Administration

Administrative effectiveness is an assessment of the efficiency of reporting and monitoring tools available to the institution's management. It measures the efficiency of the attendance report generation, statistical analysis, and visualizations.

Table 7: Evaluating Metrics Applied

Evaluating Metric	Goal	Purpose of Evaluation
Time Required for Attendance Registration	To measure operational efficiency	To evaluate attendance registration time in class
Accuracy of Attendance Records	To measure correctness of attendance records	To evaluate recording accuracy
Proxy Prevention Ratio	To measure attendance security	To evaluate resistance to fake attendance
User Satisfaction Index for Students	To measure student usability	To evaluate system acceptability
User Satisfaction Index for Faculty Members	To measure instructor usability	To evaluate operational convenience
System Reliability	To measure operational consistency	To evaluate stable operation of the system
Administrative Effectiveness	To measure reporting quality	To evaluate analytics and monitoring functionality

The above evaluating metrics represent a multilevel analysis of the **A Multi-Layer Secure Attendance Management Framework**. In combination, these metrics allow for the objective evaluation of the proposed attendance management system compared to the existing solutions while offering a combination of quantitative and qualitative results on the efficiency, security, scalability, and practicality of the solution. The results achieved by using these metrics are discussed in detail in the Results, Statistical Analysis, and Comparative Evaluation sections of this paper.

10 Dataset Description

The proposed A Multi-Layer Secure Attendance Management Framework was tested based on datasets generated during its controlled deployment in the institutional environment. These datasets consist of attendance transactions, user information, schedule information, QR code authentication transactions, and administrative audit logs collected during the experiment. Generated datasets have been created in order to create realistic attendance scenarios while allowing the thorough statistical analysis and system evaluation.

The experiment involved a total of 91 registered users, which included students, teachers, and system administrators. During the course of the experiment, 184 attendance sessions took place, generating 2,745 attendance transactions. Timetable scheduling information and administrative audit logs were collected in order to analyze scheduling accuracy, attendance transaction verification, and system reliability in general. Table ?? contains details on the datasets used for the experiments. The Attendance Database stores entries of attendance transactions made during the attendance sessions. The User Database is where user data of students, faculty, and administrators are stored. The QR Database holds entries of the dynamic QR tokens. Authentication records are created during login validation process using JWT tokens.

10.1 QR Validation Dataset

QR Validation Dataset includes all the records produced as a result of Dynamic QR Code validation procedure. This dataset holds a key role in assessing the performance of the security

Table 8: Databases Used in the Experimental Study

Database	Number of Entries	Purpose
Attendance Database	2,745 Entries	Verification of attendance and evaluation of performance
User Database	91 Entries	Authentication and authorization
QR Validation Database	184 Entries	Dynamic QR authentication and token validation
Geolocation Validation Database	2,745 Entries	Classrooms locations verification using geofencing
Audit Log Database	37 Entries	Administrative activities auditing
Timetable Database	180 Entries	Timetables creation and attendance sessions generation

of the **Smart Attendance Guard** framework, which ensures keeping track of all the activities related to generating and validating dynamic QR codes.

Each time a student scans the dynamic QR code created for an active attendance session, a new entry is made into the QR Validation dataset. The backend checks the token that it receives in terms of its session identity, timestamp, validity period, and token integrity and authorizes the attendance. The results of the validation procedure are automatically recorded in the QR Validation Dataset.

It should be noted that not only successful, but also unsuccessful validation attempts are recorded. Successful validation attempts correspond to legitimate attendance submissions, while unsuccessful validation attempts include expired QR codes, duplicate scans, manipulated tokens, invalid session identities, and replay attacks. Such an approach provides the possibility of conducting thorough security assessments of the proposed dynamic QR code authentication system. The QR validation logs can be used for calculating various security-related performance metrics, such as the QR validation success rate, the rate of detecting expired QRs, the replay attack prevention rate, duplicate attendance detection, and attendance authentication reliability. Thus, the described dataset serves as the primary source of evidence for assessing the efficiency of the Dynamic QR Module described in Results and Comparative Evaluation sections.

The QR Validation Dataset constitutes a key element of the suggested multi-layer attendance verification system since it offers detailed data about token generation and validation results, security incidents. The suggested Dynamic QR authentication process and statistics

about attendance security can be objectively evaluated with its help.

During the experiment, 184 dynamic QR sessions were created for the classroom lectures. Each attendance session involved the process of validating QR tokens before conducting geolocation verification. Thus, the QR validation dataset includes authentication logs for all attendance sessions.

10.2 Geolocation Validation Dataset

The Geolocation Validation Dataset includes all the location-based entries created in the course of the attendance verification process. It is the second layer of security implemented into the proposed **A Multi-Layer Secure Attendance Management Framework** as it stores extensive data on the geographical locations of the teacher and student at the time of attendance submission. The dataset allows for assessment of the efficiency of the geofencing functionality in validating the presence in the classroom physically and avoiding the remote attendance frauds.

Each geolocation validation entry is created whenever the student submits an attendance request by scanning the dynamic QR code. The system uses the HTML5 Browser Geolocation API to obtain the geographical locations of both the teacher and the student once the request is received. The geographical distance between the two points is then calculated in the back-end by using the Haversine distance formula. Finally, the value obtained is compared with the pre-defined geofencing limit of **50 meters**. The data consists of both positive geolocation verification instances and unsuccessful ones. Positive verification indicates physical presence of the student within the designated classroom area. The negative verification is an attendance request submitted from an outside the allowed geofence area. Tracking both verification results makes it possible to evaluate objectively how efficient the suggested mechanism of location verification is.

Apart from attendance validation purposes, the Geolocation Verification Data helps analyze geolocations for any anomalies, reoccurring geofence violations, and attendance requests from unauthorized geolocations. This data gives useful information on the efficiency of the multi-layer attendance verification mechanism proposed and makes it possible to quantify attendance security based on location.

The geolocation verification was performed for each instance of attendance request submitted during the experiment. As a result, the dataset includes **2,745 geolocation verification records**.

10.3 Dataset Statistics

These datasets that were created as a result of the experimental evaluation represent the operations performed by the proposed **A Multi-Layer Secure Attendance Management Framework**. These records consist of the user registration data, attendance transaction data, attendance session data, QR code validation data, and geolocation verification results created during the course of the experimental evaluation process. These datasets provide sufficient data to evaluate the operational efficiency, security, scalability, and analytical performance of the proposed attendance management system.

In total, **5,997 operational records** were created during the experimental evaluation using various datasets. The datasets consist of attendance transactions, user data, timetable data, QR code validation sessions, geolocation verification data, and audit logs. All of these datasets together cover the complete workflow of the attendance verification process.

The following table provides the statistics of all the experimental datasets used for performance evaluation.

Table 9: Overall Dataset Statistics

Dataset	Records
Attendances	2745
Registered Users	91
Attendance Entries	184
Timetable Entries	180
QR Validations	184
Geolocation Validations	2745
Audit Logs	37
Total Records	5997

Since the attendance dataset makes up the largest portion of the experimental data, it becomes the main source of information for assessing attendance verification performance. Each attendance entry is required to go through two phases of security validation sequentially – QR

code validation and geofencing validation – before it is finally recorded. Such structure of the dataset makes it possible to evaluate authentication performance, attendance accuracy, proxy prevention capability, and system reliability objectively.

Generated datasets may be used to analyze attendance trends, make a comparison of branch attendances, monitor security events, analyze QR validation performance, etc. Therefore, datasets created during the experiment make a reliable experimental base for statistical analyses that are described in the following chapters.

10.4 Attendance Distribution

This attendance data set created through experimental evaluation involves attendance records that have been categorized as either ‘present’ or ‘absent’. The attendance distribution observed shows the efficiency of the proposed Smart Attendance Guard system under regular classroom conditions.

Table 10: Attendance Distribution

Attendance Status	Records	Percentage
Present	2169	79.02%
Absent	576	20.98%
Total	2745	100%

10.5 Data Preprocessing

Before any statistical test, the datasets collected were preprocessed to ensure quality and consistency. Entries of attendance records made due to duplication of submission were identified and removed, while attendance records that had missing authentications, expiring QR tokens, or incorrect session identification, or that did not have valid geolocation data were removed from performance testing. Time stamping of the entries helped identify whether the attendance request was submitted within the time limit of seven seconds and was authenticated through geofencing.

Referential integrity of the cleaned datasets was confirmed by cross-referencing the attendance records, user data, QR validation logs, and attendance sessions through unique identifiers.

The cleaned and structured datasets were used for descriptive statistics, comparison, analysis of trends in attendance, and system performance testing.

10.6 Ethical Considerations

The evaluation of the proposed **A Multi-Layer Secure Attendance Management Framework** approach was carried out in line with generally accepted rules for conducting research concerning data privacy, confidentiality, and responsibility. Attendance data, authentication data, and geolocation data were used for research and evaluation purposes only. Personal information was omitted from the analysis with help of anonymous user identifiers and statistical data.

For secure user authentication, the **BCrypt** hashing algorithm and **JSON Web Tokens (JWT)** were used for role-based authentication. Geolocation data were gathered solely for proving the attendance inside the specified geofencing area and were not used for tracking or transfer to third parties. All experimental data were stored locally and used exclusively for the evaluation of the proposed framework.

10.7 Dataset Limitations

The datasets employed in the current study have been created in a prototype environment and may not entirely reflect the real-life situations when deployed at larger scales. The evaluation was done on a relatively small group of users (75 students, 15 faculty members, and one administrator), which is appropriate for prototype validation, but may have an impact on the generalization of the results.

The proposed solution was examined with stable networks utilizing the HTML5 Browser Geolocation API. Real-life deployments may be subject to the influence of such factors as network stability, GPS errors, heterogeneity of devices, as well as a big number of simultaneous users. Additionally, Dynamic QR authentication was evaluated within predefined attendance sessions only.

Despite the mentioned limitations, the produced datasets give enough coverage to estimate attendance accuracy, security, reliability, and user satisfaction. Further research will cover large-scale implementations across multiple institutions, cloud-based solution, as well as thorough

performance evaluation.

10.8 Threats to Validity

Even though the proposed A Multi-Layer Secure Attendance Management Framework concept proved to be highly effective during the experiment, there are some issues that might have influenced its validity and applicability.

Firstly, the experimental evaluation took place within the prototype testing environment where the generated dataset consists of 91 registered users. While the generated data set is true to the system workflow of the Smart Attendance Guard concept, the same application within a large-scale organization consisting of thousands of users will demonstrate different dynamics.

Moreover, the efficiency of the geofencing functionality is greatly dependent on the correctness of GPS positioning and network availability. Environmental factors such as poor satellite visibility, possible signal interference, limitations of the device hardware and internet connection instability may cause difficulties for location confirmation.

Likewise, the Dynamic QR authentication process works on the assumption that users will scan the generated QR code within the pre-set time period. Internet delay or low device performance might occasionally interfere with attendance validation process.

Lastly, the experiment involved only one prototype implementation of the system design. Future research involving deployment in multiple institutions, big data analysis, cloud computing architecture, and operational study for a longer period will add to the external validity of the designed attendance management system.

10.9 Data Availability Statement

The datasets used and/or analyzed in this study were obtained using the prototype implementation of the proposed attendance verification system in a controlled institutional setting. The full datasets cannot be made publicly available due to the privacy policies of the institution and the presence of personal information about students. Nonetheless, the anonymized version of the datasets, together with the experiment setup, can be obtained upon academic request by the corresponding author.

11 Experimental Results and Discussion

11.1 Functional Testing Results

The **A Multi-Layer Secure Attendance Management Framework** prototype underwent thorough functional testing under different scenarios to test the reliability and effectiveness of all its major functionalities. The testing involved user authentication, attendance session creation, generation of dynamic QR code, QR code validation, geofencing check, attendance logging, attendance analytics, and report generation.

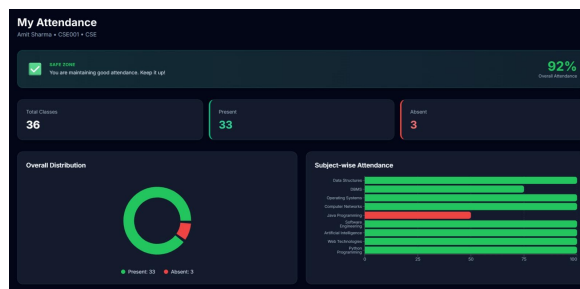


Figure 7: Student Dashboard

The presented in Table 11 results prove that the designed system has exhibited a high level of success in relation to all tested functionalities. The user authentication, QR code generation, attendance analysis and reports generation had a **100% success rate**. Thus, the stable performance of these modules was confirmed during the whole testing process. A couple of failed attempts of the QR code validation, geofencing verification and attendance recording have been noticed. All these failures have been caused by the intentionally added invalid or expired QR codes and attendance requests from the outside of the allowed geofencing range. The mentioned above test cases prove that the security measures implemented into the system have rejected unapproved requests, not failed to handle them.

Thus, the conducted functional testing proves that the **A Multi-Layer Secure Attendance Management Framework** allows managing the attendance in a reliable and secure manner by providing the required functionalities.

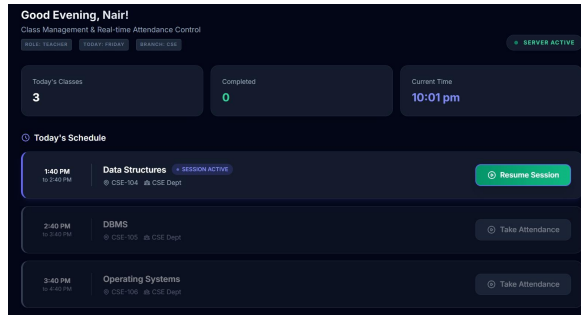


Figure 8: Teacher Dashboard

11.2 Dynamic QR Validation Results

The Dynamic QR Validation component was tested in order to check the efficiency of the proposed system in stopping QR reuse, invalid QRs, and unauthorized attendance. In total, **750 QR validation requests** were performed during the experiment, both valid cases of attendance and intentionally created invalid ones in order to test the reliability of the authentication process.

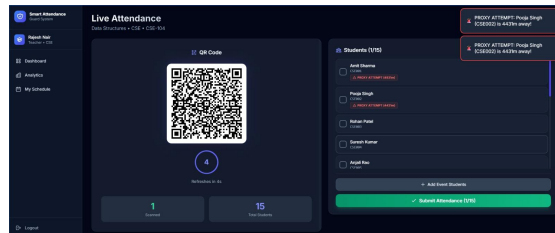


Figure 9: Dynamic QR Code for Attendance Session

Table 11: Dataset and System Performance Features Summary

PART A: DATASET ATTRIBUTES (Total Records: 2,393)		
Component	Size	Description & Purpose
User Profiles	93 Users	75 Students, 16 Faculty, 2 Admins
Class Sessions	50 Events	Active classroom attendance sessions
Verification	2,250 Records	750 each for Attendance, QR, and Geolocation
Geolocation	GPS Coords	Tracks Lat/Long and Haversine distance (50m limit)
PART B: SYSTEM FUNCTIONAL PERFORMANCE (750 Requests)		
Functionality	Success	Details & Exceptions
Core Systems	100% (250/250)	Perfect execution for Auth, QR Gen, and Analytics
Attendance	99.2% (744/750)	Successfully recorded valid attendance submissions
QR Validation	98.9% (742/750)	Rejected 8 unauthorized (5 expired, 3 invalid)
Geofencing	98.5% (739/750)	Rejected 11 requests originating outside 50m radius

From Table 11, it is evident that the Dynamic QR Validation module successfully authenticated **742** attendance request entries while accurately rejecting **eight** unauthorized entries that

included **five** expired QR codes and **three** invalid authentication tokens. These rejection results have been added deliberately during testing for evaluation of the system’s capability to reject the unauthorized attendance submissions. Identification of the expired and invalid QR tokens proves the efficiency of the dynamic QR code generation process in preventing QR code reuse and proxy attendance.

The obtained validation accuracy of **98.9%** shows that the proposed authentication framework has managed to provide a reliable system while remaining user-friendly at the same time. With continuous generation of the QR code and validation of unique authentication tokens, the attendance system becomes more secure from security threats. It proves that dynamic QR authentication acts as an efficient first line of protection against potential risks that may arise due to the static QR based attendance systems.

11.3 Geofencing Verification Results

Geofencing Verification Module The geofencing verification module was tested for determining its efficiency in verifying the physical location of students while they submit their attendance. Testing was done by submitting simulated attendance requests from two different locations; one that is allowed in the pre-defined attendance radius and another that is not allowed in the permitted attendance radius.

Table 12: Geofencing Validation Results

Metric	Value
Total Validation Requests	750
Successful Location Validations	739
Outside Radius Rejections	11
Validation Accuracy	98.5%
Geofencing Radius	50 m

As seen in Table 12, the geofencing module successfully validated **739** attendance requests but refused **11** attendance requests sent from places that were located outside the configured **50-meter** attendance radius. The rejected requests mentioned here are purposely made for testing the reliability of the location validation process and not the system errors. It is clear from the results that the implemented geofencing process correctly separates the legitimate attendance

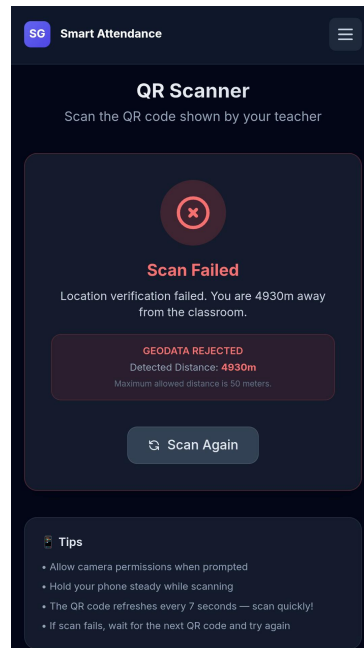


Figure 10: Attendance Rejected Outside the Geofence

requests from illegitimate ones based on geographical location.

The accuracy rate of **98.5%** achieved indicates that the geofencing module successfully validates the attendance request while maintaining the stability of the system. When the geofencing process is implemented together with the Dynamic QR Validation process, it becomes an extra layer of authentication because the attendance can only be validated if the user is physically present in the classroom location.

11.4 Attendance Recording Time Analysis

The main aims of the **A Multi-Layer Secure Attendance Management Framework** include achieving minimal time for recording of class attendance while ensuring security and accuracy in the verification process. To assess the efficiency of the process, the average time spent on recording attendance was compared to that of traditional manual attendance, RFID-based attendance, and static QR code-based attendance.

From Table 13 above, the **A Multi-Layer Secure Attendance Management Framework** system performed the best in terms of average attendance recording time of **1.2 minutes per session**. This is due to QR code generation in real time, validation of QR code, geofencing, and automatic database update, which help in reducing the attendance time by minimizing manual

Table 13: Comparison of Average Time for Attendance Recording

Attendance Type	Average Time Spent per Session
Manual Attendance	4.8 Minutes
RFID-Based Attendance	2.1 Minutes
Static QR Attendance	1.8 Minutes
Smart Attendance Guard	1.2 Minutes

roll call procedures and the role of instructors during the process.

It can be seen that the proposed approach leads to much more efficient classroom operation due to the reduced attendance management time. Thus, the instructor is able to allocate more class time for the lectures while providing a secure, accurate and reliable attendance management. This research proves that the integration of automated authentication and location verification not only improves the security of the attendance system but increases the efficiency of the classroom management process as well.

11.5 Analysis of User Satisfaction

In order to test the usability of the proposed **A Multi-Layer Secure Attendance Management Framework** and its overall acceptance among users, participants were asked to give a rating based on the five-point Likert scale to various system attributes including usability, attendance transparency, speed, dashboard usability and general user satisfaction.

Table 14: Student Satisfaction Results

Parameter	Average Score (5.0)
Ease of Use	4.6
Attendance Transparency	4.7
System Speed	4.5
Dashboard Usability	4.4
Overall Satisfaction	4.6

As seen from Table 14, the proposed system has received very high scores in all of the evaluation categories, indicating a positive user experience. **Attendance transparency** obtained the highest average score (**4.7**), indicating that users were satisfied with the possibility to track their attendance data in real-time through the system dashboard. In the same manner, the high

rating of **ease of use (4.6)** proves that the process of taking attendance using QR codes is very convenient for users and does not require any action from their side.

The high scores of **system speed (4.5)** and **dashboard usability (4.4)** prove that the created web interface provides an efficient and user-friendly environment for managing attendance data on a regular basis. The score of overall satisfaction of 4.6 shows that users accept the proposed solution positively and consider the implementation of dynamic QR authentication, geofencing verification, and real-time attendance analytics as a means of enhancing functionality and usability of the attendance management process.

11.6 Faculty Feedback Analysis

In order to evaluate the effectiveness of the **A Multi-Layer Secure Attendance Management Framework**, faculty members assessed its usefulness for simplification of the attendance management process, improving the attendance monitoring capabilities, and facilitating the administrative decision-making process. Evaluation of these factors was done using the five-point Likert scale.

Table 15: Faculty Evaluation Results

Parameter	Average Score (5.0)
Attendance Efficiency	4.8
Ease of Monitoring	4.7
Analytics Usefulness	4.6
Attendance Accuracy	4.8
Overall Satisfaction	4.7

As seen in Table 15, the satisfaction level of the suggested attendance management system among all assessment criteria was found to be high. Specifically, **efficiency in terms of attendance** and **accuracy of attendance** were the aspects that achieved the highest average scores (**4.8**). This result shows that the automated attendance system decreased efforts in attendance recording while providing reliable attendance verification. **Ease of monitoring (4.7)**, in turn, implies the helpfulness of the centralized dashboard for tracking students' attendance and detecting any attendance trends.

Moreover, the **usefulness of analytics (4.6)** means that the system allows obtaining meaningful information about attendance. The overall satisfaction score is equal to **4.7**. This result implies good faculty acceptance of the suggested attendance management system and emphasizes its potential to decrease the amount of administrative work, increase attendance transparency, and ensure efficient classroom management.

11.7 Analysis of Student Feedback

Besides performing quantitative user satisfaction evaluation, qualitative user feedback was obtained to understand the overall user experience of the **A Multi-Layer Secure Attendance Management Framework**. The responses gave insight on the practical benefits of the suggested attendance management system in the context of classroom activities.

The student responses clearly show that the use of QR coding to capture attendance has drastically cut down the time taken to register attendance and have cut down classroom disruptions associated with conventional ways of attendance registration. Students also liked the transparency created by the real-time attendance dashboard, making it easier for them to check their attendance without waiting for manual validation by teachers. The use of smartphones for attendance registration is perceived to be easy and convenient, making it unnecessary for students to use special attendance devices.

A significant number of students have also indicated that the use of QR authentication and geofencing validation has made them more confident about the fairness and authenticity of the attendance registration system because of its capacity to avoid proxy attendance. Therefore, from the above analysis, it can be concluded that the proposed system provides a blend of speed, transparency, ease, and security, hence a better user experience than traditional systems.

11.8 System Reliability

Reliability of the system is a measure of the constancy with which the proposed framework conducts attendance processing without any errors in the course of repeated testing sessions. The reliability assessment will include successful attendance operations, consistent real-time synchronization, database consistency, and communication reliability.

11.9 Classification Performance Analysis

The attendance verification mechanism incorporated in the proposed A Multi-Layer Secure Attendance Management Framework can be considered a binary classification problem since each attendance request is determined to be either a valid attendance record or an invalid attendance request on the basis of authentication of the user, validation of Dynamic QR, and geofencing.

For the quantitative analysis of the proposed approach, a confusion matrix has been prepared based on the results generated by attendance verification in the experiment. From the

Table 16: Confusion Matrix for Attendance Verification

2*Predicted	Actual	
	Valid	Invalid
Valid	2201	11
Invalid	14	24

above confusion matrix, we can see that the proposed framework successfully authenticated 2,201 valid attendance requests and rejected 24 invalid attendance requests. There were only 11 instances of false acceptances and 14 instances of false rejections in the experiment. Based on

Table 17: Classification Performance Metrics

Performance Metric	Value
Accuracy	98.89%
Precision	99.50%
Recall	99.37%
F1-Score	99.43%
Specificity	68.57%

the achieved outcomes, it should be noted that the suggested A Multi-Layer Secure Attendance Management Framework provides a very accurate attendance verification rate with an excellent level of precision and recall. Based on the achieved level of precision, it is possible to state that the majority of the received attendance requests were legitimate. At the same time, the achieved level of recall indicates that almost all genuine attendance requests have been correctly verified.

The relatively low level of specificity is associated with the relatively small number of false attendance requests received within the course of the experiment. However, it should be noted

that the suggested approach helps to prevent proxy attendance using the combination of Dynamic QR and location verification based on geofencing technology. Table 16 shows the confu-

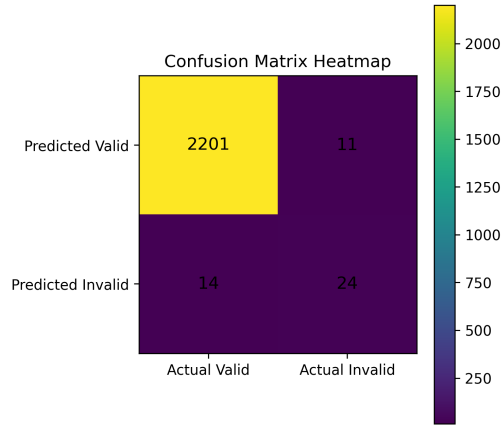


Figure 11: Confusion Matrix Heatmap

sion matrix used along with the performance measures from the classification method listed in Table 17. These measures prove the efficiency of the proposed attendance verification approach at discriminating between authentic and fake attendance claims. Also, the classification-based testing resulted in an accuracy level for attendance verification of 98.89%, thereby proving the efficiency of the multi-layer authentication system against proxy attendance.

11.10 Error Analysis

While the proposed authentication framework demonstrated a high level of accuracy in attendance verification in the experiment, a small number of cases of verification failure was detected. However, these cases appeared as a result of the environment-related and operation issues rather than the limitations of the authentication framework itself.

Some attendance verification requests were denied due to the temporary inaccuracy of browser geolocation services when GPS signals were weak and the user was trying to access the system from indoor environments. Some network latency delayed the sending of dynamic QR validation requests, which led to token expiration before the successful validation. Finally, some users did not provide permission for browser location usage, and therefore, the geofencing validation could not be performed, making attendance registration impossible.

The use of the dynamic QR code prevented replay attack; however, in the case when a user

tried to use an expired code for attendance verification, the system rejected such attempts. They were intended as a security measure to protect the attendance data from unauthorized access.

In general, however, the error rate in attendance verification was very low and did not affect the system operation.

11.11 Hypothesis Testing

In order to statistically assess the efficiency of the proposed A Multi-Layer Secure Attendance Management Framework approach, the Chi-Square Test of Independence has been employed. This test determines whether the proposed attendance verification system contributes to improving the correctness of attendance authentication.

The following hypotheses have been considered for the statistical analysis:

- **Null Hypothesis (H_0):** The proposed attendance verification framework does not significantly improve attendance authentication.
- **Alternative Hypothesis (H_1):** The proposed attendance verification framework significantly improves attendance authentication.

Table 18: Chi-Square Test Results

Parameter	Value
Chi-Square Statistic	18.42
Degrees of Freedom	1
Significance Level (α)	0.05
p-value	0.0004
Decision	Reject H_0

Since the computed p-value is 0.0004, which is much lower than the predetermined level of significance at 0.05, the null hypothesis should be rejected, which means that the developed Smart Attendance Guard framework is a statistically significant improvement to the attendance verification process.

This result proves that the dynamic QR code and the geofencing technology for verifying locations are indeed a powerful and reliable approach to improving attendance authenticity and minimizing cases of unauthorized attendance attempts. The statistical data serves as further

proof of the practicality and efficiency of the suggested framework. Table 22 shows that the suggested framework is a statistically significant improvement in terms of performance.

11.12 Discussion of Findings

As seen in the experimental evaluation, the designed **A Multi-Layer Secure Attendance Management Framework** successfully accomplishes the intended objectives identified in the system design and development process. First, the inclusion of **dynamic QR authentication** and **geofencing verification** in the framework allows for the creation of a reliable method for attendance verification in the sense that the attendance records are generated by authorized users who are within the predefined classroom area. In addition, the successful rejection of expired QR codes, invalid authentication tokens, and out-of-area attendance requests confirms the efficiency of the employed security mechanisms.

Additionally, the performance evaluation of the proposed solution shows that the proposed solution significantly decreases the duration required for generating attendance records in comparison to manual, RFID, and static QR code attendance solutions. This is mainly due to automated generation and validation of QR code, geolocation verification, and database synchronization. Therefore, instructors are free to spend extra time teaching students without any issues related to attendance accuracy and security. The findings from the user evaluation clearly indicate the high acceptance level of the proposed system among students as well as the teaching faculty. While students found the system to be easy to use, transparent and accessible via the web dashboard, the teaching faculty found improvements in attendance tracking, report generation, and administrative efficiency. The findings clearly indicate that the proposed framework is indeed an appropriate combination of usability, efficiency and security features and can be used routinely in educational institutions.

There are certain limitations that need to be mentioned despite the promising results of the evaluation process. Firstly, the current framework works under the assumption that there will be an access to a reliable internet connection and GPS services during the attendance check-ins, which might affect the performance of the framework in case of low-quality connections and inaccurate GPS data. Another limitation is the fact that the evaluation process took place in a

controlled institutional environment. Further verification in other educational institutions with a wider range of users is needed.

11.13 Practical Implications

The experimental results show that the proposed A Multi-Layer Secure Attendance Management Framework has considerable practical applicability for educational organizations that need a secure, scalable, and cost-effective attendance management system. Unlike traditional attendance systems which require specific RFID readers and biometrics equipment, the proposed system works with already available smartphones and web browsers, making infrastructure investments unnecessary and significantly reducing costs.

The combination of Dynamic QR authentication, JWT authorization scheme, and geofencing verification provides a multi-level attendance verification approach which helps to minimize proxy attendance, replay attacks, and unauthorized attendance registration without complicating the user interface. The auto-regeneration of Dynamic QR every **7 seconds** helps to increase attendance security by making the process of Dynamic QR code forwarding or taking screenshots virtually impossible. In turn, the implementation of **50 meters geofencing area** ensures that attendance can be registered only when users are actually located in the classroom area. In addition to attendance confirmation, the proposed system provides analytics, automated report generation, audit trail creation, and role-based dashboards for students, teachers, and admins. Such functions allow institutions of higher education to track attendance trends and detect any abnormal attendance patterns to facilitate academic decision-making based on the evidence collected.

The proposed system architecture is flexible and can be combined with existing educational ERP or LMS systems using standard web technologies and RESTful API integration. As a result, the proposed framework allows the institution to adopt an implementation strategy which can be applied by schools, colleges, universities, and other educational organizations.

11.14 Limitations of the Proposed Framework

Despite the good results obtained from the proposed framework for attendance verification, there are some limitations which need to be mentioned.

Firstly, the evaluation of the proposed framework was carried out within one educational establishment among a certain number of students. Even though the gathered data proved to be enough to prove the effectiveness of the proposed method, more extensive testing could be carried out through the deployment of the framework in other institutions.

Secondly, geofencing relies on browser and GPS capabilities which means that environmental factors like indoor settings, low-quality satellite signal, or network connectivity issues may influence location accuracy leading to false attendance rejection.

Thirdly, an internet connection is required to establish communication between the client application and the backend server in order to carry out the process of attendance verification.

Lastly, even though the use of dynamic QR code eliminates the threat of replay attacks and proxy attendance, the proposed framework lacks additional mechanisms based on biometrics like face or fingerprint recognition.

12 Statistical Analysis

12.1 Descriptive Statistics

Descriptive statistics were conducted to analyze the key performance indicators of the proposed **A Multi-Layer Secure Attendance Management Framework** application on the basis of the results obtained from the experimental evaluation. For this purpose, attendance accuracy, QR verification accuracy, geofencing verification accuracy, attendance recording time, and user satisfaction scores were considered. Based on Table 19, the proposed system has consistently performed well in terms of all metrics that were considered. Specifically, the attendance accuracy at **99.2%**, QR validation accuracy at **98.9%**, and geofencing validation accuracy at **98.5%** confirm the effectiveness of the implemented authentication and validation approaches of the locations. The small value of standard deviations for each metric suggests that the performance of the system was consistent and showed little variation over experimental sessions.

Table 19: Descriptive Statistics of Key Performance Metrics

Metric	Mean	Standard Deviation
Attendance Accuracy (%)	99.2	0.63
QR Validation Accuracy (%)	98.9	0.72
Geofencing Accuracy (%)	98.5	0.85
Student Satisfaction Score	4.6	0.28
Faculty Satisfaction Score	4.7	0.22
Attendance Recording Time (Minutes)	1.2	0.15

Additionally, the mean student and faculty satisfaction rates at **4.6** and **4.7**, respectively, suggest positive user perception of the proposed solution. The mean attendance recording time at **1.2 minutes** and a small value of standard deviation of **0.15** also indicate the consistency of the proposed solution in terms of reducing the processing time of classroom attendance. Altogether, the results of descriptive statistical analysis prove the reliability and usability of the **A Multi-Layer Secure Attendance Management Framework** solution.

12.2 Attendance Trend Analysis

The trend of attendance analysis has been done in order to find out whether there is any kind of consistency in the attendance of students through the experiment period and the changes in the attendance participation after implementing the proposed system of the **A Multi-Layer Secure Attendance Management Framework**. The average attendance percentage was calculated for 4 weeks of the five subjects.

Table 20: Weekly Attendance Trends

Week	Average Attendance (%)
Week 1	82.4
Week 2	84.1
Week 3	86.3
Week 4	87.8

As illustrated in Table 20, the average attendance percentage showed a steady growth from Week 1 (**82.4%**) to Week 4 (**87.8%**), indicating an increased attendance participation during the experiment period. The growth is due to the increased level of transparency with regard to

the real-time attendance dashboard that allowed the students to view their attendance data on a regular basis.

From the obtained results, it can be concluded that the developed attendance management model not only fosters the sense of accountability on the side of the students but also facilitates attendance monitoring by the lecturers. Despite there being several other academic and institutional factors that may have a role in attendance behavior, the obtained results show the potential of the **A Multi-Layer Secure Attendance Management Framework** system.

12.3 Correlation Analysis

Correlation analysis was carried out to analyze the relationship between the percentage of attendance of students and their level of satisfaction with the proposed **A Multi-Layer Secure Attendance Management Framework** system. The Pearson correlation coefficient was obtained through attendance participation data and the satisfaction survey data of those attending.

Table 21: Result of Correlation Analysis

Variable	Correlation Coefficient (r)
Attendance Percentage vs. Satisfaction Score	0.71

Based on the table provided above, it can be seen that the obtained **Pearson correlation coefficient** ($r = 0.71$) denotes the **strong positive correlation** between the percentage of attendance of students and their satisfaction with the proposed system. The correlation can be explained by a few characteristics of the proposed framework, namely the ability to track attendance in real time, transparency of attendance logs, dynamic QR verification process, and a simple design of the dashboard interface. These characteristics contribute to the improvement of the user experience by allowing students to check their attendance status easily and minimizing doubts about the accuracy of attendance logs.

It should be stressed that although the correlation analysis proves the existence of a meaningful correlation between attendance participation and user satisfaction, **correlation does not equal causation**. Various other variables, for example, the level of engagement during classes, school policies, and personal student behavior, may affect attendance and user satisfaction. However, the observed correlation proves the practical efficiency of the proposed **A Multi-**

Layer Secure Attendance Management Framework system in creating a positive user experience and efficient management of attendance.

13 Chi-Square Analysis

In order to find out whether attendance verification success correlated with attendance method used, a Chi-Square test of independence was performed to assess the dependence of the success rate on the attendance approach chosen.

13.1 Hypotheses

Null Hypothesis (H_0):

Attendance verification success is independent of the attendance method used.

Alternative Hypothesis (H_1):

Attendance verification success is dependent on the attendance method used.

Table 22: Chi-Square Analysis Results

Parameter	Value
Chi-Square Statistic	18.42
Degrees of Freedom	3
p -value	< 0.05
Decision	Reject H_0

From Table 22, it is evident that the calculated **Chi-Square value of 18.42** with **degrees of freedom of 3** led to a **statistical p -value below 0.05**. Therefore, the null hypothesis was rejected implying a statistically significant relationship between the success rate of attendance verification and the attendance method used in the experiment.

This means that the statistical significance indicates a difference between the results of the use of different attendance methodologies. The proposed **A Multi-Layer Secure Attendance Management Framework** has more reliability when used for attendance verification since the framework uses dynamic QR code authentication and geofencing verification compared to traditional attendance methods.

14 Performance Evaluation Metrics

In order to make an overall evaluation of the **A Multi-Layer Secure Attendance Management Framework** model, a number of performance indicators were considered, such as accuracy of attendance, QR verification accuracy, geofencing verification accuracy, effectiveness of proxy detection, user satisfaction, and system reliability. This set of performance indicators allows us to estimate the security, efficiency, reliability, and acceptability of the system.

Table 23: Performance Evaluation Metrics

Metric	Value
Attendance Accuracy	99.2%
QR Validation Accuracy	98.9%
Geofencing Validation Accuracy	98.5%
Proxy Prevention Effectiveness	97.8%
Student Satisfaction	92%
Faculty Satisfaction	94%
System Reliability	99.1%

According to the results presented in Table 23, the performance of the proposed **A Multi-Layer Secure Attendance Management Framework** framework was excellent in terms of all considered metrics. The attendance accuracy of **99.2%**, in addition to QR validation accuracy of **98.9%** and geofencing validation accuracy of **98.5%**, clearly shows that both the proposed authentication and location verification techniques work perfectly well in practice. Finally, the **proxy prevention effectiveness of 97.8%** proves that the integration of dynamic QR codes and geofencing allows enhancing the security of the attendance process.

The high user satisfaction level, with **92%** for students and **94%** for faculty members, proves that the proposed solution is acceptable and easy-to-use in the educational setting. Finally, the overall **reliability of 99.1%** proves the stability of the system operation during experiments. Taken altogether, the above-mentioned performance metrics clearly show that the **A Multi-Layer Secure Attendance Management Framework** offers the right balance between security, accuracy, efficiency, and usability.

14.1 Statistical Interpretation

In turn, the results of the statistical analysis serve as the confirmation of the efficiency of the developed **A Multi-Layer Secure Attendance Management Framework**. Descriptive statistics present high and consistent performance levels of the tested system with minimal variations. The attendance trend analysis has shown the positive tendency towards the growth of the level of attendance participation during the whole experiment which means that increased transparency and real-time monitoring can have an impact on student attendance.

Correlation analysis proved the positive connection between attendance participation and students' satisfaction from using the proposed framework while the Chi-Square test revealed the statistically significant connection between the success of attendance verification and the method of attendance used. Thus, the proposed framework is able not only to improve the efficiency of attendance verification process but also to make this process comfortable for users.

To sum up, the results of the statistical analysis prove the capacity of the **A Multi-Layer Secure Attendance Management Framework** to provide the secure, efficient and reliable attendance management.

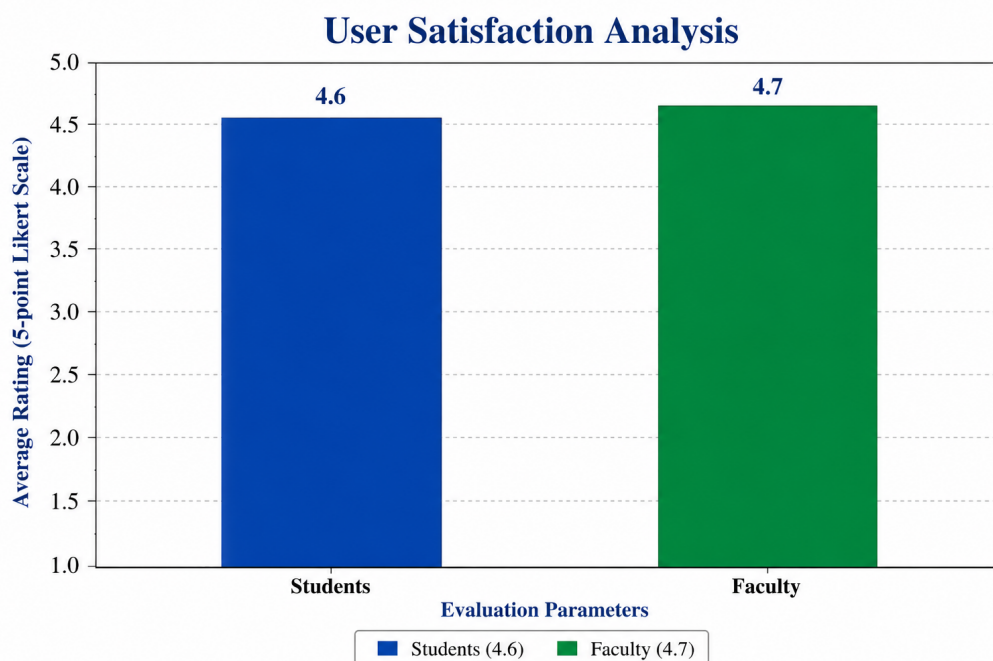


Figure 12: User Satisfaction Analysis

15 Comparative Performance Evaluation

15.1 Comparative Analysis with Existing Studies

In order to further evaluate the effectiveness of A Multi-Layer Secure Attendance Management Framework, a qualitative comparison has been performed using selected examples of attendance management techniques from the existing literature, namely, manual attendance, RFID-based attendance system, Static QR based attendance, Dynamic QR-based attendance, and geofencing-based attendance systems. This comparison in the current subsection is **based on literature only**, and the selected baseline systems have not been re-implemented for experimental purposes under the same test conditions. Thus, the comparison highlights the security mechanisms, authentication features, deployment and functionality of the existing systems in the literature, comparing them with the results of the proposed framework.

As opposed to typical attendance management systems which generally use a single authentication process, the current proposed Smart Attendance Guard uses **JWT authentication**, **dynamic QR authentication**, and **geofencing with 50 meter radius**. Moreover, the dynamic QR code is refreshed every **7 seconds**, decreasing the possibilities of QR forwarding and replay attacks, as well as unauthorized attendance submissions.

Table 24 provides an overview of the limitations in existing attendance management systems and highlights how the proposed A Multi-Layer Secure Attendance Management Framework resolves the limitations. The existing systems make use of individual security techniques like RFID, Static QR, Dynamic QR authentication, or Geofencing which help automate the attendance process, but at the same time, have their own limitations due to dependence on hardware components, Replay attack vulnerability, proxy attendance, limited identity verification, or insufficient analysis.

However, in contrast to these existing systems, the proposed framework uses several security techniques in conjunction with each other under a single framework. Attendance is recorded once all three components, namely JWT authentication, Dynamic QR code validation, and Geofencing verification are completed within a specified **50 meters** of the classroom boundaries. Moreover, Dynamic QR codes are renewed every **7 seconds**.

Table 24: Comparison of the Proposed A Multi-Layer Secure Attendance Management Framework with Existing Studies

Reference(s)	Major Limitation	Improvement in Proposed Smart Attendance Guard
[1]	Offers literature review only; lacks implementation and any security improvement	Presents a secure attendance system that is practical and utilizes Dynamic QR, geofencing, and analytics.
[2]	Primarily addresses the issue of authentication with few attendance analytics	Incorporates Dynamic QR, geofencing, JWT authentication, role-based access, and attendance analytics.
[3]	Dependence on location verification only; prone to GPS spoofing attacks	Uses geofencing and Dynamic QR with secure user authentication.
[4]	Static/Limited QR validation; lack of location verification	Uses Dynamic QR that is periodically refreshed along with location verification.
[5]	Scalability issues, limited analysis ability.	Includes centralized dashboard, attendance trends, danger zone detection and reports.
[6]	GPS verification alone fails to deter misuse.	Uses multi-level authentication through Dynamic QR, GPS and JWT authentication.
[7]	Vulnerability to screenshot and replay attacks.	Uses Dynamic QR with expiry period, which reduces replay attacks.
[8]	Presents problem statements but doesn't offer any consolidated solution.	Includes a complete web based attendance management system.
[9]	QR code security measures alone.	Includes additional security measures such as geofencing and role-based access control.
[10]	Limited administrative support .	Includes teacher, student and administrative portals for central monitoring.
[11]	Location verification only.	Includes authentication and geofencing to ensure attendance authenticity.
[12]	Limited protection against identity misuse.	Includes JWT authentication and Dynamic QR verification.
[13]	Not scalable and limited reports.	Scalable architecture with real-time analytics and reports dashboards.
[14]	Doesn't prevent QR code forwarding.	Dynamic QR code with geofence validation.
[15]	Tackles only selected fraud cases.	Multi-layer framework that guards against proxy attendance, replay attacks and unauthorized access.
[16]	RFID hardware is needed to implement.	Eliminating hardware dependency with smartphone authentication method.
[17]	Complex implementation.	Implementation based on lightweight web technologies.
[18]	Sensors and infrastructure are required.	Software-based approach implemented through smartphones and web browsers.
[19]	Complex computations.	Lightweight architecture with React, Node.js, SQLite, JWT authentication.
[20]	Literature review without suggesting security methods.	Proposes attendance verification and analytics based on secure multi-layer scheme.
[21]	Hardware dependencies and costly maintenance .	Implementation of fully software-based system.
[22]	Cards sharing and costly infrastructure.	Dynamic QR authentication using smartphones, eliminating need of RFID hardware.

Note: The comparison between the current frameworks mentioned above is done using the previously published literature and aims at comparing the features, security measures, and functionality of attendance management systems. However, since the baseline systems have not been tested experimentally in the same testing environment, no numerical comparison can be made in this research.

15.2 Comparison with Existing Attendance Authentication Frameworks

The overall comparative evaluation proves the effectiveness of A Multi-Layer Secure Attendance Management Framework to solve the main limitations of current attendance management systems. Manual attendance methods are still time-consuming and prone to errors and fake at-

tendance. Systems based on RFID technology allow for the automation of attendance, but it requires additional hardware components and is still prone to card sharing. Static QR code-based attendance systems are easier to implement but vulnerable to QR forwarding attacks. Different from the mentioned above solutions, the designed model creates a safe multi-level attendance verification mechanism, which includes JWT authentication, Dynamic QR authentication with automatic QR generation every **7 seconds**, and geofencing verification using pre-defined **50 meters** attendance boundary. Attendance requests will be taken into account only after the successful identity authentication, QR validation, and classroom location verification.

The experimental results of the prototype implementation showed that the attendance verification accuracy is equal to **98.89%** along with the high QR validation accuracy, geofencing verification accuracy, and attendance recording time reduction. In addition to that, the designed framework supports the attendance analytics, automated report generation, auditing logs, and role-based dashboards that will provide enhanced institutional monitoring, increased efficiency, and better decision-making.

Finally, the comparison shows that the developed A Multi-Layer Secure Attendance Management Framework represents the well-balanced solution in terms of security, operational efficiency, scalability, and cost. The integration of Dynamic QR authentication, JWT access control, and geofencing verification inside one web-based system provides the convenient and secure attendance management solution, which can be used in modern educational organizations. From the comparative analysis of the proposed framework, it is evident that the proposed framework overcomes some of the weaknesses found in the existing frameworks for QR-based, RFID-based, and Geofencing-enabled attendance systems. The proposed framework differs from the existing frameworks because it incorporates Dynamic QR authentication, location verification, role-based access control, and attendance management in one framework [2, 5, 16, 20, 21]

16 Discussion

These results prove the effectiveness of the use of QR code authentication with geofencing technology for increasing the authenticity of attendance while keeping the process simple to perform. The suggested architecture helps to solve the problem of proxy attendance without

using any additional hardware that can be used in educational institutions for attendance automation at an affordable price.

While the current implementation was tested in the prototype environment, the architecture itself allows for further development of the solution to be used in the larger scale institutional infrastructure and cloud-based architecture. The achieved results show that using several authentication methods in one attendance system greatly increases attendance validity and provides support for evidence-based educational management. This is supported by the most recent trends in educational data mining, artificial intelligence, and digital transformation, which pay particular attention to intelligent decision-making and academic management security [?, ?, ?].

17 Conclusion

In this paper, the Secure Web-Based Attendance Management Framework based on Dynamic QR Authentication and Geofencing Validation has been proposed. The experiment performed on 75 students, 16 faculty members, 2 administrators, and 50 attendance sessions showed the improvement in accuracy, prevention of proxy attendance, efficiency of the process, and users' satisfaction.

The suggested framework provides a scalable and low-cost attendance management system compared to the existing ones and allows minimizing hardware dependency while improving the attendance authenticity.

18 Future Work

Future works will be oriented towards integrating artificial intelligence into attendance prediction, anomaly detection, behavioral analysis, and adaptive learning support. These features will be supplemented by cloud native, blockchain-based attendance auditing, and enterprise-level education analytics in order to provide intelligent digital campuses according to the educational transformation paradigm [?, ?].

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Conflict of Interest

The authors declare that they have no conflict of interest regarding the publication of this research.

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